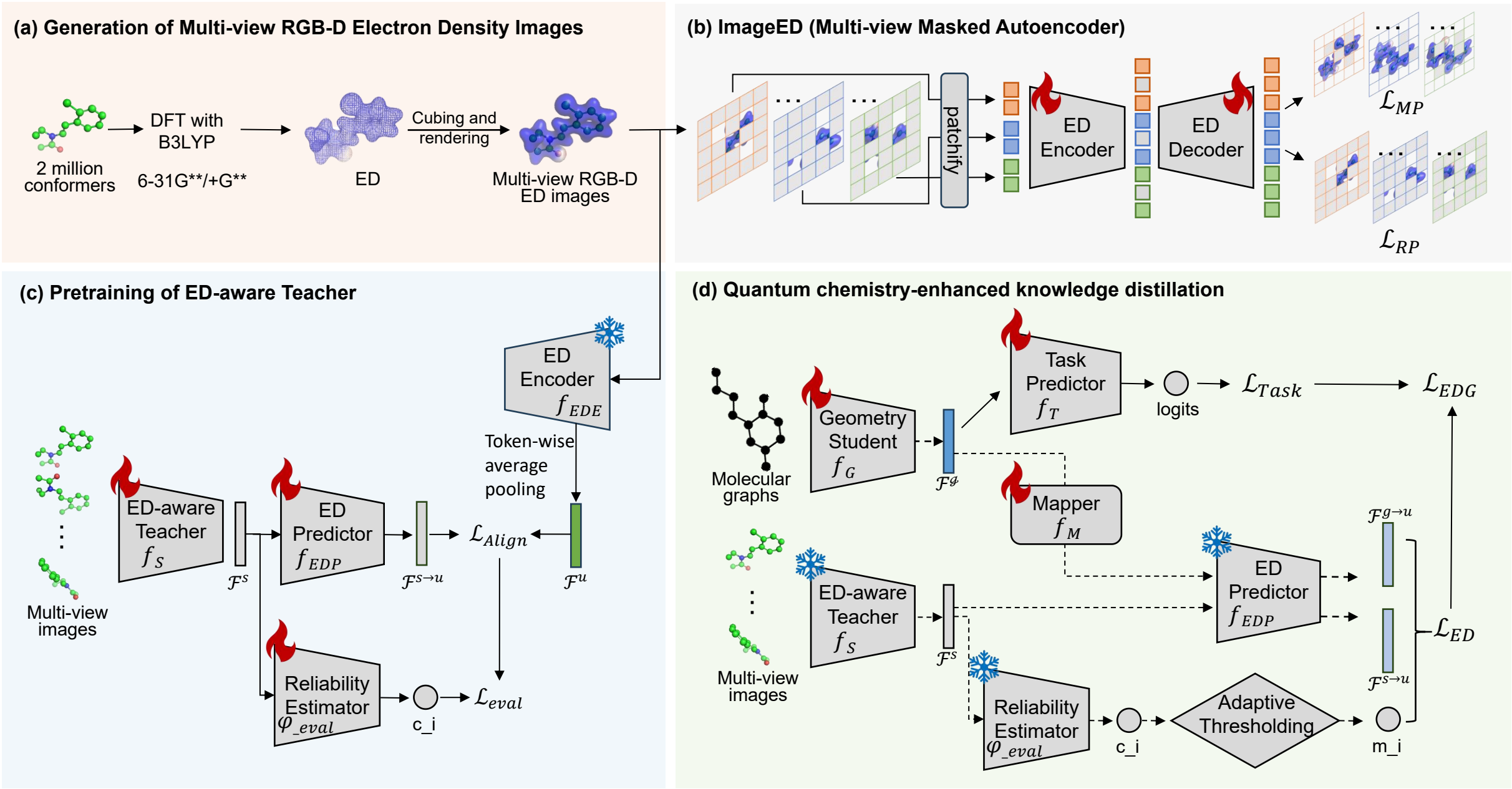
(a) Generation of Multi-view RGB-D Electron Density Images
2 million conformers
DFT with B3LYP
6-31G**/+G**
ED
Cubing and rendering
Multi-view RGB-D ED images
(b) ImageED (Multi-view Masked Autoencoder)
patchify
ED Encoder
ED Decoder
$\mathcal{L}_{MP}$
$\mathcal{L}_{RP}$
(c) Pretraining of ED-aware Teacher
ED Encoder f_{EDE}
Token-wise average pooling
Multi-view images
ED-aware Teacher f_S
$\mathcal{F}^s$
ED Predictor f_{EDP}
$\mathcal{F}^{s\to u}$
$\mathcal{L}_{Align}$
$\mathcal{F}^u$
Reliability Estimator φ_eval
c_i
$\mathcal{L}_{eval}$
(d) Quantum chemistry-enhanced knowledge distillation
Molecular graphs
Geometry Student f_G
$\mathcal{F}^g$
Task Predictor f_T
logits
$\mathcal{L}_{Task}$
$\mathcal{L}_{EDG}$
Mapper f_M
Multi-view images
ED-aware Teacher f_S
$\mathcal{F}^s$
ED Predictor f_{EDP}
$\mathcal{F}^{g\to u}$
$\mathcal{F}^{s\to u}$
$\mathcal{L}_{ED}$
Reliability Estimator φ_eval
c_i
Adaptive Thresholding
m_i